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|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
|  |                                                                                                                                                                                                                                                                                                                                        | <b>Parul University</b><br><b>Faculty of Engineering and Technology</b><br><b>Parul Institute of Engineering and Technology</b><br><b>Department of Artificial Intelligence and Data Science</b> |                |
| <b>Subject Name</b>                                                               | <b>Theory of Computations</b>                                                                                                                                                                                                                                                                                                          | <b>A.Y</b>                                                                                                                                                                                       | <b>2026-27</b> |
| <b>Subject Code</b>                                                               | <b>303105306</b>                                                                                                                                                                                                                                                                                                                       | <b>Semester</b>                                                                                                                                                                                  | <b>5</b>       |
| <b>Research Article/Papers for Theory of Computation</b>                          |                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                  |                |
| <b>SR. NO.</b>                                                                    | <b>Link</b>                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                  |                |
| <b>1</b>                                                                          | <b>The Complexity of Small Universal Turing Machines: A Survey</b><br>Excellent survey paper on Universal Turing Machines and computational complexity.<br><a href="https://www.sciencedirect.com/science/article/pii/S0304397508007287?via%3Dihub">https://www.sciencedirect.com/science/article/pii/S0304397508007287?via%3Dihub</a> |                                                                                                                                                                                                  |                |
| <b>2</b>                                                                          | <b>Undecidability and Finite Automata</b><br>Research paper discussing undecidability problems related to finite automata.<br><a href="https://arxiv.org/abs/1702.01394">https://arxiv.org/abs/1702.01394</a>                                                                                                                          |                                                                                                                                                                                                  |                |
| <b>3</b>                                                                          | <b>On the Computability of Graph Turing Machines</b><br>Advanced paper exploring graph-based Turing machine models and computability.<br><a href="https://arxiv.org/abs/1703.09406">https://arxiv.org/abs/1703.09406</a>                                                                                                               |                                                                                                                                                                                                  |                |
| <b>4</b>                                                                          | <b>From Undecidability of Non-Triviality and Finiteness to Undecidability of Learnability</b><br>Connects undecidability theory with modern machine learning concepts.<br><a href="https://arxiv.org/abs/2106.01382">https://arxiv.org/abs/2106.01382</a>                                                                              |                                                                                                                                                                                                  |                |
| <b>5</b>                                                                          | <b>Did Turing Prove the Undecidability of the Halting Problem?</b><br>Modern discussion and analysis of Turing's undecidability proof.<br><a href="https://academic.oup.com/logcom/article/36/1/exaf075/8417148">https://academic.oup.com/logcom/article/36/1/exaf075/8417148</a>                                                      |                                                                                                                                                                                                  |                |